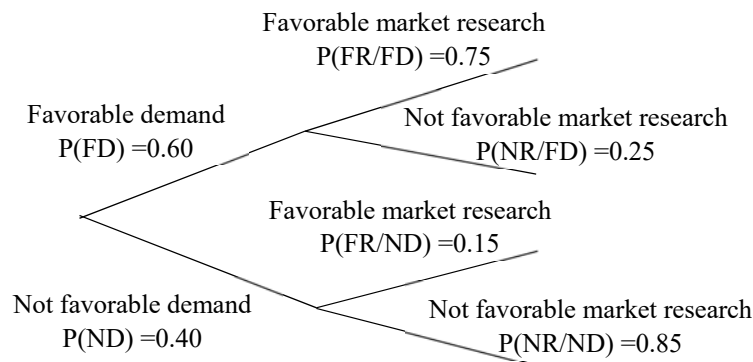


Decision Modelling (NPTEL Online Course)

Assignment 2: Decision Analysis

Detailed Answers

- 1) An electronic gadget manufacturing enterprise needs to decide whether to invest in a new product development proposal. Estimates based on preliminary knowledge available show the probability of favorable demand $P(\text{FD})$ to be 0.60. In order to be more assured, the company can conduct a market research. If the demand is favorable, market research becomes favorable with a probability of 0.75. If the demand is not favorable, market research becomes unfavorable with a probability of 0.85 (See the figure below):



From the Figure above, probability of getting favorable demand given the market research is favorable, i.e. $P(\text{FD} | \text{FR})$ will be:

- i. Joint probability
 - ii. Prior probability
 - iii. **Posterior probability**
 - iv. Total or unconditional probability
- 2) The probability of favorable market research (FR) given that there favorable demand, i.e. $P(\text{FR} | \text{FD})$ will be (Refer Figure in Question 1):
- i. Prior probability
 - ii. Posterior probability
 - iii. **Conditional probability**
 - iv. Joint probability

3) Which of the following expressions gives the value of total or unconditional probability for favorable market research $P(\text{FR})$ (refer Figure in question 1):

- i. $P(\text{FR}) = P(\text{FR}|\text{FD}) * P(\text{FD}) + P(\text{NR}|\text{FD}) * P(\text{FD})$
- ii. **$P(\text{FR}) = P(\text{FR}|\text{FD}) * P(\text{FD}) + P(\text{FR}|\text{ND}) * P(\text{ND})$**
- iii. $P(\text{FR}) = P(\text{NR}|\text{FD}) * P(\text{FD}) + P(\text{FR}|\text{ND}) * P(\text{ND})$
- iv. $P(\text{FR}) = P(\text{FR}|\text{FD}) * P(\text{FD}) + P(\text{NR}|\text{ND}) * P(\text{ND})$

4) Posterior Probability in the Bayesian Analysis, is a type of conditional probability. This is:

- i. **True**
- ii. False
- iii. Not certain

5) In Bayesian analysis:

- i. **Joint probabilities are obtained by multiplying prior probabilities with appropriate conditional probabilities**
- ii. Joint probabilities are obtained by dividing prior probabilities with appropriate conditional probabilities
- iii. Joint probabilities are obtained by adding prior probabilities with appropriate conditional probabilities
- iv. None of these

6) Consider the following two statements about decision trees and identify the correct statements from the options that follow:

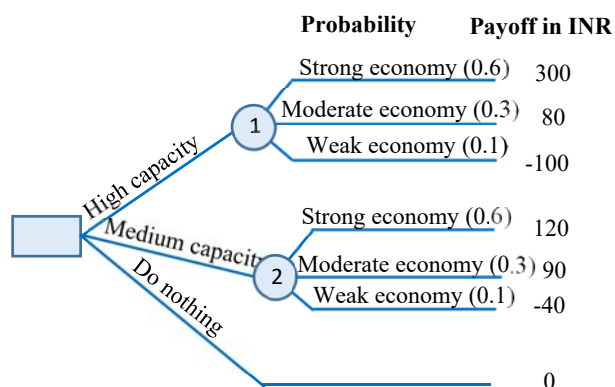
Statement 1: a state-of-nature node is shown using a circle

Statement 2: decision alternatives are shown using arcs

- i. Only Statement 1 is true
- ii. Only statement 2 is true
- iii. **Both the statements are true**
- iv. Neither of the statements is true

7) For the decision tree given below, what will be the right decision for a decision maker who operates with the expected value criterion?

- i. **High capacity**
- ii. Medium capacity
- iii. Do nothing
- iv. Cannot be decided



Answer to Q7

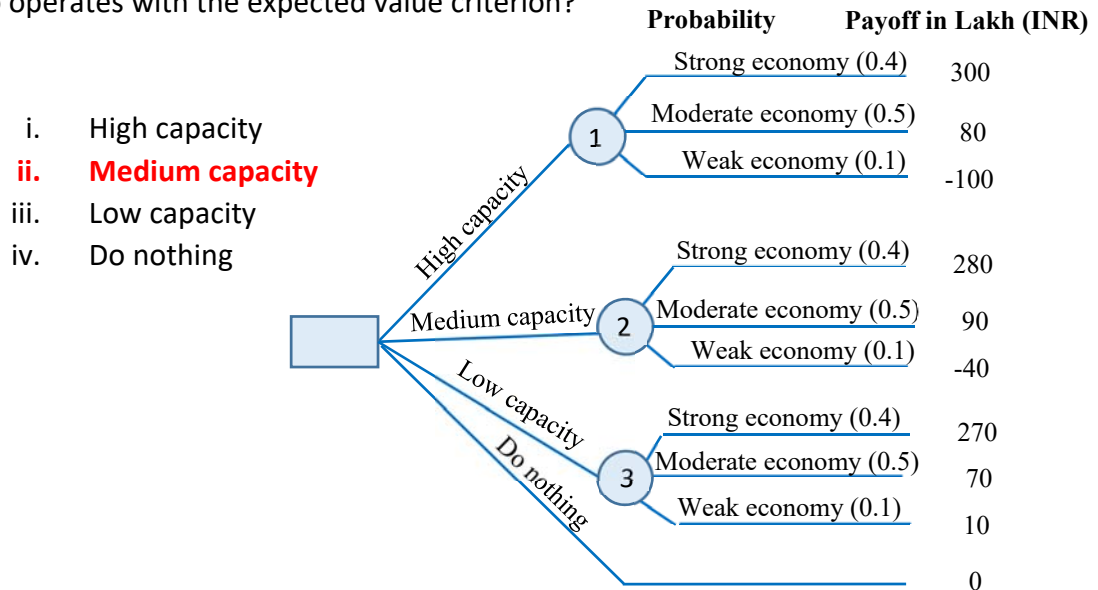
Payoff for High capacity = $0.6 \times 300 + 0.3 \times 80 + 0.1 \times (-100) = 194$

Payoff for Medium capacity = $0.6 \times 120 + 0.3 \times 90 + 0.1 \times (-40) = 95$

Payoff for 'do nothing' = 0

Therefore, the decision maker will choose **High capacity**

- 8) For the decision tree given below, what will be the right decision for a decision maker who operates with the expected value criterion?



Answer to Q8

Payoff for High capacity = $0.4 \times 300 + 0.5 \times 80 + 0.1 \times (-100) = 150$

Payoff for Medium capacity = $0.4 \times 280 + 0.5 \times 90 + 0.1 \times (-40) = 153$

Payoff for Low capacity = $0.4 \times 270 + 0.5 \times 70 + 0.1 \times 10 = 144$

Payoff for 'do nothing' = 0

Therefore, the decision maker will choose **Medium capacity**

- 9) For the data given below, what is the right decision for the expected value decision rule

Alternatives	State of nature (probability)		
	Most favorable (0.3)	Moderately favorable (0.5)	Unfavorable (0.2)
Strategy 1	1500	800	-300
Strategy 2	1200	900	-150
Strategy 3	1300	1100	-100
Strategy 4	1250	1000	-50

- i. Strategy 1
- ii. Strategy 2
- iii. Strategy 3**
- iv. Strategy 4

Answer to Q9.

Calculation of payoffs using expected value decision rule:

- Payoff for Strategy 1 = $1500 \times 0.3 + 800 \times 0.5 + (-300) \times 0.2 = 790$
- Payoff for Strategy 2 = $1200 \times 0.3 + 900 \times 0.5 + (-150) \times 0.2 = 780$
- Payoff for Strategy 3 = $1300 \times 0.3 + 1100 \times 0.5 + (-100) \times 0.2 = \mathbf{920}$
- Payoff for Strategy 4 = $1250 \times 0.3 + 1000 \times 0.5 + (-50) \times 0.2 = 865$

From payoff calculations the selected alternative is **Strategy 3**.

10) For decision making with experimentation which of the following is correct?

- i. The cost of experimentation does not affect the decision alternative payoffs
- ii. The cost of experimentation affects the decision alternative payoffs**
- iii. The cost of experimentation affects the decision alternative payoffs only when such experiments alter the posterior probabilities
- iv. The cost of experimentation affects the decision alternative payoffs only when such experiments alter the final decision made

11) A company needs to decide whether to develop a new product or go with the existing one. The management knows the initial payoffs. Now the company can go for a market research to find the market acceptability of the new product, which modifies the prior probabilities to a set of posterior probabilities. The first decision now is whether to go for a market research or not. The second decision is whether to develop the new product or not. In this situation which of the following statements is correct:

- i. The decision of developing the new product is dependent on the outcome of the market research**
- ii. The decision of market research is dependent on the decision of the new product development
- iii. Both the decisions are independent of each other
- iv. None of these.

12) A company needs to decide whether to develop a new product or go with the existing one. The management knows the initial payoffs. Now the company can go for a market research to find the market acceptability of the new product, which modifies the prior probabilities to a set of posterior probabilities. The first decision now is whether to go for a market research or not. The second decision is whether to develop the new product or not. For this situation, consider the two statements identify the correct option that follow:

Statement 1: When market research is favorable, posterior probability of market acceptability will be higher than its prior probability

Statement 2: When market research is not favorable, posterior probability of market acceptability will be lesser than its prior probability

- i. Only Statement 1 is true
- ii. Only statement 2 is true
- iii. Both of the statements are true**
- iv. Neither of the statements are true

13) In a decision making situation with experimentation, Expected Value of Best Decision without Sample Information is Rs. 30000. With sample information, the expected value of Best Decision goes up to Rs. 35000. The sample information is available at a cost of Rs. 5000. Expected Value of Sample Information (EVSI) is:

- i. Rs. 0
- ii. Rs. 5000
- iii. Rs. 10000**
- iv. Rs. 30000

Answer to Q13

Expected Value of Sample Information (EVSI) =

Expected value with sample information, along with the cost to gather it –

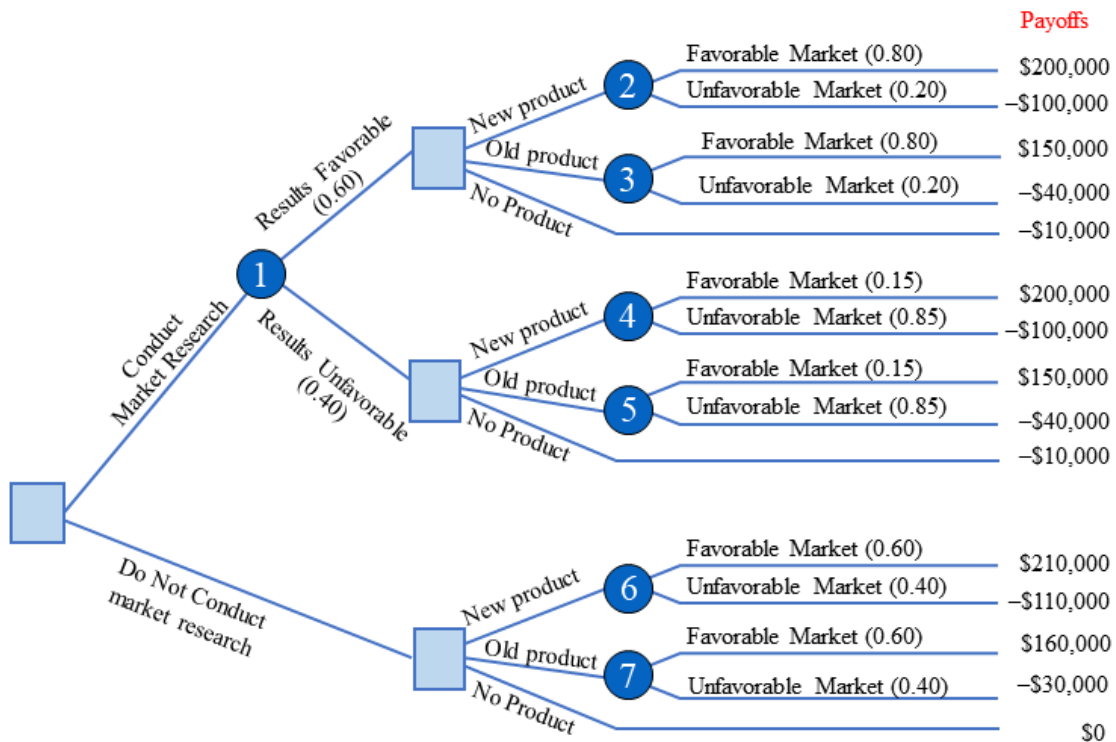
Expected value of best decision without sample information

This actually gives the actual value of conducting the survey to gather the sample information.

So, Expected value of sample information (EVSI) = $(35000 + 5000) - 30000 = 10000$

14) A company is having the following decision tree to make decision related to development of a new product. If the market research gives favorable results, the company should

- i. **Go with the new product**
- ii. Go with the old product
- iii. Should avoid releasing any products



Answer to Q14.

Expected value calculation for nodes

Given favorable market research results (Results favorable)

EV (node 2) = EV (new product | Results favorable)

$$= (0.80) * (\$200,000) + (0.20) * (-\$100,000) = \mathbf{\$140,000}$$

EV (node 3) = EV (old product | Results favorable)

$$= (0.80) * (\$150,000) + (0.20) * (-\$40,000) = \$112,000$$

If no product is released the payoff is **-\$10,000**

If the market research results are favorable, company should **go with the new product** with expected value of **\$140,000**

15) In Question 14, if the market research result is unfavorable, the company should:

- i. Go with the new product
- ii. Go with the old product
- iii. Avoid releasing any products**
- iv. Cannot be decided with certainty

Answer to Q15.

Expected value calculation for nodes

Given unfavorable market research results (Results unfavorable)

EV (node 4) = EV (new product | Results favorable)

$$= (0.15) * (\$200,000) + (0.85) * (-\$100,000) = -\$55,000$$

EV (node 5) = EV (old product | Results favorable)

$$= (0.15) * (\$150,000) + (0.85) * (-\$40,000) = -\$11,500$$

If no product is released the payoff is **-\$10,000**

If the market research results are unfavorable, company **should avoid releasing any products** to invite minimum loss.

16) In a city dangerous fires are rare (1%) but smoke is fairly common (10%) due to barbecues.

If 90% of dangerous fires make smoke, what is the probability of dangerous fire when there is smoke?

- i. 7%
- ii. 8%
- iii. 9%**
- iv. 10%

Answer to Q16.

$P(\text{Fire}) = 1\%$; $P(\text{Smoke}) = 10\%$

$P(\text{Smoke} | \text{Fire}) = 90\%$

Hence, Posterior Probability $P(\text{Fire} | \text{Smoke}) = [P(\text{Fire}) * P(\text{Smoke} | \text{Fire})] / P(\text{Smoke})$
 $= [(0.01 * 0.9)] / 0.1 = 9\%$

17) In a city the probability of rain in the evening is 5% according to historical data for a particular season and 60% of the time when there is a rain, it leads to traffic jam. If in that city traffic jam probability is 10%, and in an evening there is a traffic jam, what is the probability that it has rained?

- i. 20%
- ii. 25%
- iii. 30%**
- iv. 35%

Answer to Q17.

$P(\text{rain}) = 5\%$; $P(\text{traffic-jam}) = 10\%$

$P(\text{traffic-jam} \mid \text{rain}) = 60\%$

$P(\text{rain} \mid \text{traffic-jam}) = [P(\text{rain}) * P(\text{traffic-jam} \mid \text{rain})] / P(\text{traffic-jam})$
 $= [(0.05 * 0.6)] / 0.1 = 0.3 = \mathbf{30\%}$

18) For a particular brand of computers, if it has hard disk problems 80% of the times the startup is slow. As per manufacturer's description, within warranty period, the probability of a slow startup is 2% whereas the probability of hard disk problem is 0.5%. If the computer is starting slow in the warranty period, what is the probability that the hard disk has developed problems?

- i. 20%**
- ii. 25%
- iii. 30%
- iv. 35%

Answer to Q18.

$P(\text{slow}) = 2\%$; $P(\text{HD problem}) = 0.5\%$

$P(\text{slow} \mid \text{HD problem}) = 80\%$

$P(\text{HD problem} \mid \text{slow}) = [P(\text{HD problem}) * P(\text{slow} \mid \text{HD problem})] / P(\text{slow})$
 $= [(0.5\% * 80\%)] / 2\% = \mathbf{20\%}$

19) A testing procedure for a certain disease correctly gives positive report 98% of the time and correctly gives negative report 96% of the time. If the disease has infected 1% of the population and one randomly selected person is tested positive, what is the probability that he is infected by the disease?

- i. 18.9%
- ii. **19.8%**
- iii. 22.7%
- iv. 22.8%

Answer to Q19.

Let, a person is infected be represented as A

Result positive be represented by B

Given,

$$P(A) = 1\% = 0.01 \Rightarrow P(\sim A) = 0.99$$

$$P(B|A) = 98\% = 0.98, \Rightarrow P(\sim B|A) = 0.02$$

$$P(\sim B|\sim A) = 96\% = 0.96, \Rightarrow P(B|\sim A) = 0.04$$

Probability that the person is infected if the test result is positive:

$$P(A|B) = P(B|A) \cdot P(A) / [P(B|A) \cdot P(A) + P(B|\sim A) \cdot P(\sim A)]$$

$$= (0.98 \cdot 0.01) / [(0.98 \cdot 0.01) + (0.04 \cdot 0.99)]$$

$$= (0.0098) / (0.0098 + 0.0396)$$

$$= 0.198 = \mathbf{19.8\%}$$

20) In a company four investment options are being evaluated that depends on the market condition. what is the right decision for the expected value decision rule considering the following data?

Alternatives	State of nature (probability)		
	Most favorable (0.4)	Moderately favorable (0.35)	Unfavorable (0.25)
Option 1	1800	1200	-250
Option 2	1750	1250	-200
Option 3	1300	1400	-100
Option 4	1250	1500	-150

- i. Option 1
- ii. **Option 2**
- iii. Option 3
- iv. Option 4

Answer to Q20.

	State of nature (probability)			Payoff calculations
Alternatives	Most favorable (0.4)	Moderately favorable (0.35)	Unfavorable (0.25)	
Option 1	1800	1200	-250	$1800 * 0.4 + 1200 * 0.35 + (-250) * 0.25$ $= 1077.5$
Option 2	1750	1250	-200	$1750 * 0.4 + 1250 * 0.35 + (-200) * 0.25$ $= \mathbf{1087.5}$
Option 3	1300	1400	-100	$1300 * 0.4 + 1400 * 0.35 + (-100) * 0.25$ $= 985$
Option 4	1250	1500	-150	$1250 * 0.4 + 1500 * 0.35 + (-150) * 0.25$ $= 987.5$

So, the choice would be **Option 2** for expected value criteria.